

**LAPORAN PRAKTIKUM
PEMROGRAMAN WEB 2
MODUL 5**



FUNCTION & DATABASE

Oleh:

ERISCHA MARSELA

NIM. 2410817120022

**PROGRAM STUDI TEKNOLOGI INFORMASI
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LEMBAR PENGESAHAN
LAPORAN PRAKTIKUM PEMROGRAMAN WEB 2
MODUL 4

Laporan Praktikum Pemrograman Web 2 Modul 5: FUNCTION & DATABASE ini disusun sebagai syarat lulus mata kuliah Praktikum Pemrograman Web II. Laporan Praktikum ini dikerjakan oleh:

Nama Praktikan : ERISCHA MARSELA
NIM : 2410817120022

Menyetujui,
Asisten Praktikum

Mengetahui,
Dosen Penanggung Jawab Praktikum

Ghani Mudzakir
NIM. 2310817110011

Ir. Muhammad Alkaff, S.Kom., M.Kom.
Ph.D.
NIP. 19860613 201504 1 001

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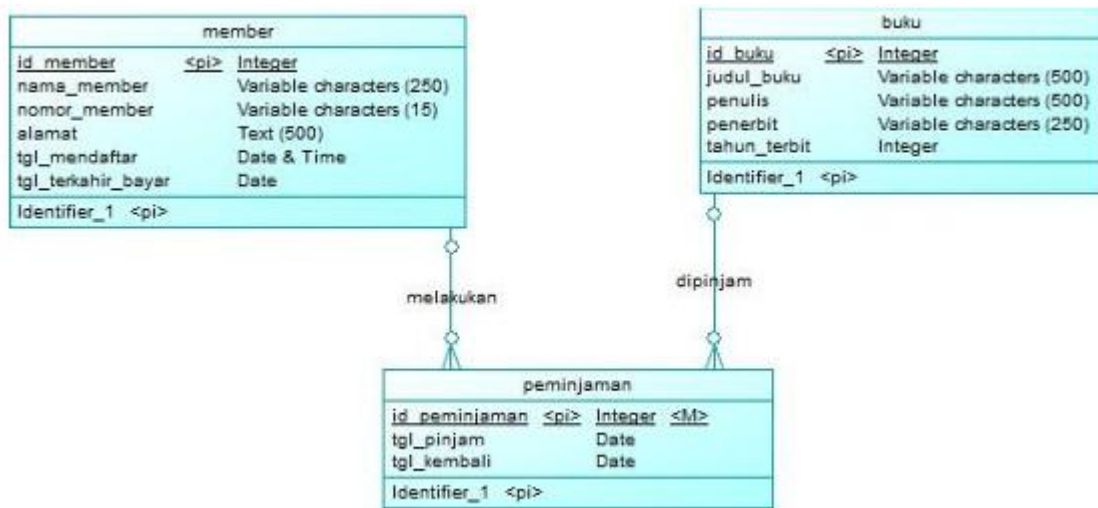
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SOAL 1

Diberikan desain konseptual sebuah basis data seperti berikut :



Gambar 1. Soal nomor 1

Buatlah sebuah aplikasi berbasis web sederhana yang dapat melakukan operasi CRUD (Create, Read, Update, Delete) dari hasil implementasi desain basis data yang diberikan. Adapun ketentuan pembuatannya sebagai berikut:

1. Koneksi database dibuat menjadi satu file sendiri yaitu Koneksi.php, kemudian gunakan fungsi require ketika ingin melakukan operasi ke basis data.
2. Operasi data seperti Insert, Update, Delete, Get Data dibuat menjadi fungsi sendiri masing-masing dan disimpan di dalam satu file khusus yaitu Model.php
3. Minimal terdapat file-file berikut di dalam proyek

Tabel 1. Contoh file yang harus ada

NAMA FILE	KETERANGAN
Koneksi.php	Berisi fungsi-fungsi untuk melakukan koneksi ke basis data
Model.php	Berisi fungsi-fungsi untuk melakukan operasi data seperti insert, update, delete, get data untuk semua tabel
Member.php	Menampilkan semua data member dalam bentuk tabel dan terdapat tombol untuk melakukan tambah, edit, dan hapus data member.
Buku.php	Menampilkan semua data buku dalam bentuk tabel dan terdapat tombol untuk melakukan tambah, edit dan hapus data buku.
Peminjaman.php	Menampilkan semua data peminjaman dalam bentuk tabel dan terdapat tombol untuk melakukan tambah, edit, dan hapus data peminjaman.
FormMember.php	Menampilkan formulir isian untuk menambah atau mengubah data member.

FormBuku.php	Menampilkan formulir isian untuk menambah atau mengubah data buku.
FormPeminjaman.php	Menampilkan formulir isian untuk menambah atau mengubah data peminjaman.

Simpan dengan nama folder: PRAK501

A. Source Code

Tabel 2. Source Code Koneksi.php

1.	<?php
2.	define('DB_HOST', 'localhost');
3.	define('DB_USER', 'root');
4.	define('DB_PASS', '');
5.	define('DB_NAME', 'perpustakaan');
6.	
7.	function koneksiDatabase() {
8.	\$koneksi = mysqli_connect(DB_HOST, DB_USER, DB_PASS,
	DB_NAME);
9.	
10.	if (!\$koneksi) {
11.	die("Koneksi gagal: " . mysqli_connect_error());
12.	}
13.	
14.	mysqli_set_charset(\$koneksi, 'utf8');
15.	return \$koneksi;
16.	}

Tabel 3. Source Code Model.php

1.	<?php
2.	require_once 'Koneksi.php';
3.	
4.	function getAllMember() {
5.	\$koneksi = koneksiDatabase();
6.	\$query = "SELECT * FROM member ORDER BY id_member ASC";
7.	\$result = mysqli_query(\$koneksi, \$query);
8.	\$data = [];
9.	while (\$row = mysqli_fetch_assoc(\$result)) {
10.	\$data[] = \$row;
11.	}
12.	mysqli_close(\$koneksi);
13.	return \$data;
14.	}
15.	
16.	function getMemberById(\$id_member) {
17.	\$koneksi = koneksiDatabase();
18.	\$id = mysqli_real_escape_string(\$koneksi,
	\$id_member);
19.	\$query = "SELECT * FROM member WHERE id_member = '\$id'
	LIMIT 1";
20.	\$result = mysqli_query(\$koneksi, \$query);
21.	\$data = mysqli_fetch_assoc(\$result);
22.	mysqli_close(\$koneksi);

```

23.     return $data;
24. }
25.
26. function insertMember($nama_member, $nomor_member, $alamat,
    $tgl_mendaftar, $tgl_terakhir_bayar) {
27.     $koneksi          = koneksiDatabase();
28.     $nama_member       = mysqli_real_escape_string($koneksi,
    $nama_member);
29.     $nomor_member      = mysqli_real_escape_string($koneksi,
    $nomor_member);
30.     $alamat            = mysqli_real_escape_string($koneksi,
    $alamat);
31.     $tgl_mendaftar     = mysqli_real_escape_string($koneksi,
    $tgl_mendaftar);
32.     $tgl_terakhir_bayar =
    mysqli_real_escape_string($koneksi, $tgl_terakhir_bayar);
33.
34.     $tgl_bayar_val     = ($tgl_terakhir_bayar !== '') ?
    "$tgl_terakhir_bayar" : "NULL";
35.
36.     $query = "INSERT INTO member (nama_member, nomor_member,
    alamat, tgl_mendaftar, tgl_terakhir_bayar)
37.         VALUES ('$nama_member', '$nomor_member',
    '$alamat', '$tgl_mendaftar', $tgl_bayar_val)";
38.
39.     $result = mysqli_query($koneksi, $query);
40.     mysqli_close($koneksi);
41.     return $result;
42. }
43.
44. function updateMember($id_member, $nama_member,
    $nomor_member, $alamat, $tgl_mendaftar, $tgl_terakhir_bayar)
    {
45.     $koneksi          = koneksiDatabase();
46.     $id_member         = mysqli_real_escape_string($koneksi,
    $id_member);
47.     $nama_member       = mysqli_real_escape_string($koneksi,
    $nama_member);
48.     $nomor_member      = mysqli_real_escape_string($koneksi,
    $nomor_member);
49.     $alamat            = mysqli_real_escape_string($koneksi,
    $alamat);
50.     $tgl_mendaftar     = mysqli_real_escape_string($koneksi,
    $tgl_mendaftar);
51.     $tgl_terakhir_bayar =
    mysqli_real_escape_string($koneksi, $tgl_terakhir_bayar);
52.
53.     $tgl_bayar_val     = ($tgl_terakhir_bayar !== '') ?
    "$tgl_terakhir_bayar" : "NULL";
54.
55.     $query = "UPDATE member
56.         SET nama member = '$nama_member',

```



```

57.             nomor_member      = '$nomor_member',
58.             alamat             = '$alamat',
59.             tgl_mendaftar      = '$tgl_mendaftar',
60.             tgl_terakhir_bayar = $tgl_bayar_val
61.         WHERE id_member = '$id_member';
62.
63.     $result = mysqli_query($koneksi, $query);
64.     mysqli_close($koneksi);
65.     return $result;
66. }
67.
68. function deleteMember($id_member) {
69.     $koneksi = koneksiDatabase();
70.     $id       = mysqli_real_escape_string($koneksi,
71.     $id_member);
72.     $query    = "DELETE FROM member WHERE id member = '$id'";
73.     $result   = mysqli_query($koneksi, $query);
74.     mysqli_close($koneksi);
75.     return $result;
76. }
77. function getAllBuku() {
78.     $koneksi = koneksiDatabase();
79.     $query    = "SELECT * FROM buku ORDER BY id_buku ASC";
80.     $result   = mysqli_query($koneksi, $query);
81.     $data     = [];
82.     while ($row = mysqli_fetch_assoc($result)) {
83.         $data[] = $row;
84.     }
85.     mysqli_close($koneksi);
86.     return $data;
87. }
88.
89. function getBukuById($id_buku) {
90.     $koneksi = koneksiDatabase();
91.     $id       = mysqli_real_escape_string($koneksi, $id_buku);
92.     $query    = "SELECT * FROM buku WHERE id_buku = '$id'
93.     LIMIT 1";
94.     $result   = mysqli_query($koneksi, $query);
95.     $data     = mysqli_fetch_assoc($result);
96.     mysqli_close($koneksi);
97.     return $data;
98. }
99. function insertBuku($judul_buku, $penulis, $penerbit,
100. $tahun_terbit) {
101.     $koneksi = koneksiDatabase();
102.     $judul_buku = mysqli_real_escape_string($koneksi,
103.     $judul_buku);
104.     $penulis    = mysqli_real_escape_string($koneksi,
105.     $penulis);

```

```

103     $penerbit          = mysqli_real_escape_string($koneksi,
104     $penerbit);
104     $tahun_terbit      = mysqli_real_escape_string($koneksi,
105     $tahun_terbit);
106     $query = "INSERT INTO buku (judul_buku, penulis,
107     penerbit, tahun_terbit)
107     VALUES ('$judul_buku', '$penulis',
108     '$penerbit', '$tahun_terbit')";
109     $result = mysqli_query($koneksi, $query);
110     mysqli_close($koneksi);
111     return $result;
112 }
113
114 function updateBuku($id_buku, $judul_buku, $penulis,
115 $penerbit, $tahun_terbit) {
116     $koneksi = koneksiDatabase();
117     $id_buku = mysqli_real_escape_string($koneksi,
118     $id_buku);
119     $judul_buku = mysqli_real_escape_string($koneksi,
120     $judul_buku);
121     $penulis = mysqli_real_escape_string($koneksi,
122     $penulis);
123     $penerbit = mysqli_real_escape_string($koneksi,
124     $penerbit);
125     $tahun_terbit = mysqli_real_escape_string($koneksi,
126     $tahun_terbit);
127     $query = "UPDATE buku
128     SET judul_buku = '$judul_buku',
129     penulis = '$penulis',
130     penerbit = '$penerbit',
131     tahun_terbit = '$tahun_terbit'
132     WHERE id_buku = '$id_buku'";
133     $result = mysqli_query($koneksi, $query);
134     mysqli_close($koneksi);
135     return $result;
136 }
137
138 function deleteBuku($id_buku) {
139     $koneksi = koneksiDatabase();
140     $id = mysqli_real_escape_string($koneksi, $id_buku);
141     $query = "DELETE FROM buku WHERE id_buku = '$id'";
142     $result = mysqli_query($koneksi, $query);
143     mysqli_close($koneksi);
144     return $result;
145 }
146
147 function getAllPeminjaman() {
148     $koneksi = koneksiDatabase();

```

```

145     $query      = "SELECT p.*, m.nama_member, m.nomor_member,
146     b.judul_buku
147             FROM peminjaman p
148             JOIN member m ON p.id_member = m.id_member
149             JOIN buku   b ON p.id_buku   = b.id_buku
150             ORDER BY p.id_peminjaman ASC";
151     $result      = mysqli_query($koneksi, $query);
152     $data        = [];
153     while ($row = mysqli_fetch_assoc($result)) {
154         $data[] = $row;
155     }
156     mysqli_close($koneksi);
157     return $data;
158 }
159 function getPeminjamanById($id_peminjaman) {
160     $koneksi = koneksiDatabase();
161     $id       = mysqli_real_escape_string($koneksi,
162     $id_peminjaman);
163     $query    = "SELECT p.*, m.nama_member, b.judul_buku
164             FROM peminjaman p
165             JOIN member m ON p.id_member = m.id_member
166             JOIN buku   b ON p.id_buku   = b.id_buku
167             WHERE p.id_peminjaman = '$id'
168             LIMIT 1";
169     $result    = mysqli_query($koneksi, $query);
170     $data      = mysqli_fetch_assoc($result);
171     mysqli_close($koneksi);
172     return $data;
173 }
174 function insertPeminjaman($id_member, $id_buku, $tgl_pinjam,
175 $tgl_kembali) {
176     $koneksi = koneksiDatabase();
177     $id_member = mysqli_real_escape_string($koneksi,
178     $id_member);
179     $id_buku   = mysqli_real_escape_string($koneksi,
180     $id_buku);
181     $tgl_pinjam = mysqli_real_escape_string($koneksi,
182     $tgl_pinjam);
183     $tgl_kembali = mysqli_real_escape_string($koneksi,
184     $tgl_kembali);
185     $query = "INSERT INTO peminjaman (id_member, id_buku,
186     tgl_pinjam, tgl_kembali)
187             VALUES ('$id_member', '$id_buku',
188             '$tgl_pinjam', '$tgl_kembali')";
189     $result = mysqli_query($koneksi, $query);
190     mysqli_close($koneksi);
191     return $result;
192 }

```

```

188
189 function      updatePeminjaman($id_peminjaman,      $id_member,
190 $id_buku, $tgl_pinjam, $tgl_kembali) {
191     $koneksi      = koneksiDatabase();
192     $id_peminjaman  = mysqli_real_escape_string($koneksi,
193 $id_peminjaman);
194     $id_member      = mysqli_real_escape_string($koneksi,
195 $id_member);
196     $id_buku        = mysqli_real_escape_string($koneksi,
197 $id_buku);
198     $tgl_pinjam     = mysqli_real_escape_string($koneksi,
199 $tgl_pinjam);
200     $tgl_kembali    = mysqli_real_escape_string($koneksi,
201 $tgl_kembali);
202
203     $query = "UPDATE peminjaman
204             SET id_member = '$id_member',
205               id_buku    = '$id_buku',
206               tgl_pinjam = '$tgl_pinjam',
207               tgl_kembali = '$tgl_kembali'
208             WHERE id_peminjaman = '$id_peminjaman'";
209
210     $result = mysqli_query($koneksi, $query);
211     mysqli_close($koneksi);
212     return $result;
213 }
214
215 function deletePeminjaman($id_peminjaman) {
216     $koneksi = koneksiDatabase();
217     $id       = mysqli_real_escape_string($koneksi,
218 $id_peminjaman);
219     $query    = "DELETE FROM peminjaman WHERE id_peminjaman =
220 '$id'";
221     $result   = mysqli_query($koneksi, $query);
222     mysqli_close($koneksi);
223     return $result;
224 }
225
226 function selesaikanPeminjaman($id_peminjaman) {
227     $koneksi      = koneksiDatabase();
228     $id            = mysqli_real_escape_string($koneksi,
229 $id_peminjaman);
230     $tgl_selesai   = date('Y-m-d');
231     $query = "UPDATE peminjaman
232             SET status      = 'selesai',
233               tgl_kembali = '$tgl_selesai'
234             WHERE id_peminjaman = '$id' AND status =
235 'aktif'";
236     $result = mysqli_query($koneksi, $query);
237     mysqli_close($koneksi);
238     return $result;
239 }

```

Tabel 4. Source Code Index.php

1.	<!DOCTYPE html>
2.	<html lang="id">
3.	<head>
4.	<meta charset="UTF-8">
5.	<meta name="viewport" content="width=device-width,
	initial-scale=1.0">
6.	<title>Perpustakaan Digital</title>
7.	<link
	href="https://fonts.googleapis.com/css2?family=Playfair+Dis
	play:wght@700&family=DM+Sans:wght@400;500;600&display=swap"
	rel="stylesheet">
8.	<style>
9.	:root {
10.	--cream: #F5F0E8;
11.	--brown: #3D2B1F;
12.	--gold: #C8960C;
13.	--rust: #A63D2F;
14.	--sage: #4A6741;
15.	--shadow: rgba(61,43,31,0.18);
16.	}
17.	
18.	* { margin: 0; padding: 0; box-sizing: border-box; }
19.	
20.	body {
21.	min-height: 100vh;
22.	background: var(--cream);
23.	font-family: 'DM Sans', sans-serif;
24.	display: flex;
25.	align-items: center;
26.	justify-content: center;
27.	position: relative;
28.	overflow: hidden;
29.	}
30.	
31.	body::before {
32.	content: '';
33.	position: fixed;
34.	top: 0; left: 0;
35.	width: 100%; height: 100%;
36.	background:
	url("https://img.freepik.com/premium-photo/too-many-books-
	library-library-inside-modern-library-interior-
	inside_179935-62265.jpg") center/cover no-repeat;
37.	pointer-events: none;
38.	}
39.	
40.	.card {
41.	background: #fff;
42.	border-radius: 20px;
43.	padding: 56px 64px;

```
44.         box-shadow: 0 24px 64px var(--shadow);
45.         max-width: 520px;
46.         width: 90%;
47.         text-align: center;
48.         position: relative;
49.         border-top: 6px solid var(--brown);
50.     }
51.
52.     .card::before {
53.         content: '';
54.         position: absolute;
55.         top: -6px; left: 50%;
56.         transform: translateX(-50%);
57.         width: 80px; height: 6px;
58.         background: var(--gold);
59.         border-radius: 0 0 4px 4px;
60.     }
61.
62.     .logo {
63.         font-size: 56px;
64.         margin-bottom: 16px;
65.         display: block;
66.     }
67.
68.     h1 {
69.         font-family: 'Playfair Display', serif;
70.         font-size: 2rem;
71.         color: var(--brown);
72.         line-height: 1.2;
73.         margin-bottom: 8px;
74.     }
75.
76.     .subtitle {
77.         color: #888;
78.         font-size: 0.9rem;
79.         margin-bottom: 40px;
80.         letter-spacing: 0.04em;
81.     }
82.
83.     .divider {
84.         width: 48px;
85.         height: 3px;
86.         background: var(--gold);
87.         margin: 0 auto 40px;
88.         border-radius: 2px;
89.     }
90.
91.     .menu-list {
92.         display: flex;
93.         flex-direction: column;
94.         gap: 14px;
95.     }
```

```

96.
97.     .menu-btn {
98.         display: block;
99.         text-decoration: none;
100        padding: 16px 24px;
101        border-radius: 12px;
102        font-weight: 600;
103        font-size: 1rem;
104        letter-spacing: 0.03em;
105        transition: all 0.22s ease;
106        position: relative;
107        overflow: hidden;
108    }
109
110    .menu-btn::after {
111        content: '→';
112        position: absolute;
113        right: 20px;
114        top: 50%;
115        transform: translateY(-50%) translateX(6px);
116        opacity: 0;
117        transition: all 0.22s ease;
118    }
119
120    .menu-btn:hover::after {
121        opacity: 1;
122        transform: translateY(-50%) translateX(0);
123    }
124
125    .btn-member {
126        background: var(--brown);
127        color: #fff;
128    }
129    .btn-member:hover { background: #5a3d2b; transform:
translateY(-2px); box-shadow: 0 8px 24px var(--shadow); }
130
131    .btn-buku {
132        background: var(--sage);
133        color: #fff;
134    }
135    .btn-buku:hover { background: #3a5432; transform:
translateY(-2px); box-shadow: 0 8px 24px
rgba(74,103,65,0.3); }
136
137    .btn-peminjaman {
138        background: var(--rust);
139        color: #fff;
140    }
141    .btn-peminjaman:hover { background: #8a3225;
transform: translateY(-2px); box-shadow: 0 8px 24px
rgba(166,61,47,0.3); }
142

```

143	.footer-text {
144	margin-top: 36px;
145	font-size: 0.78rem;
146	color: #bbb;
147	letter-spacing: 0.06em;
148	text-transform: uppercase;
149	}
150	</style>
151	</head>
152	<body>
153	<div class="card">
154	&img alt="Logo" data-bbox="500 268 518 282"/>
155	Perpustakaan Digital
156	<p class="subtitle">Sistem Informasi Manajemen
157	Perpustakaan</p>
158	<div class="divider"></div>
159	<nav class="menu-list">
160	<a href="Member.php" class="menu-btn btn-
161	member">Data Member
162	<a href="Buku.php" class="menu-btn btn-
163	buku">Data Buku
164	<a href="Peminjaman.php" class="menu-btn btn-
165	peminjaman">Data Peminjaman
166	</nav>
167	<p class="footer-text">Erischa Marsela — Modul
168	5</p>
	</div>
	</body>
	</html>

Tabel 5.Source Code Member.php

1.	<?php
2.	require_once 'Model.php';
3.	
4.	if (isset(\$_GET['hapus'])) {
5.	\$id = \$_GET['hapus'];
6.	deleteMember(\$id);
7.	header("Location: Member.php?pesan=hapus_berhasil");
8.	exit;
9.	}
10.	
11.	\$members = getAllMember();
12.	\$pesan = \$_GET['pesan'] ?? '';
13.	?>
14.	<!DOCTYPE html>
15.	<html lang="id">
16.	<head>
17.	<meta charset="UTF-8">
18.	<meta name="viewport" content="width=device-width,
	initial-scale=1.0">

19.	<title>Data Member - Perpustakaan</title>
20.	<link href="https://fonts.googleapis.com/css2?family=Playfair+Display:wght@700&family=DM+Sans:wght@400;500;600&display=swap" rel="stylesheet">
21.	<style>
22.	:root {
23.	--cream: #F5F0E8;
24.	--brown: #3D2B1F;
25.	--gold: #C8960C;
26.	--rust: #A63D2F;
27.	--sage: #4A6741;
28.	--light: #FAF8F4;
29.	}
30.	* { margin: 0; padding: 0; box-sizing: border-box; }
31.	
32.	body {
33.	min-height: 100vh;
34.	background: var(--cream);
35.	font-family: 'DM Sans', sans-serif;
36.	display: flex;
37.	}
38.	
39.	.sidebar {
40.	width: 15%;
41.	min-width: 250px;
42.	min-height: 100vh;
43.	background: var(--brown);
44.	padding: 40px 24px;
45.	display: flex;
46.	flex-direction: column;
47.	gap: 10px;
48.	flex-shrink: 0;
49.	box-shadow: 4px 0 24px rgba(61,43,31,0.05);
50.	}
51.	
52.	.sidebar h2 {
53.	font-family: 'Playfair Display', serif;
54.	color: #fff;
55.	font-size: 1.3rem;
56.	margin-bottom: 8px;
57.	padding-bottom: 20px;
58.	border-bottom: 1px solid rgba(255,255,255,0.15);
59.	}
60.	
61.	.sidebar a {
62.	display: block;
63.	padding: 12px 16px;
64.	color: rgba(255,255,255,0.75);
65.	text-decoration: none;
66.	border-radius: 10px;
67.	font-size: 0.95rem;

```
68.         font-weight: 500;
69.         transition: all 0.2s ease;
70.     }
71.
72.     .sidebar a:hover, .sidebar a.active {
73.         background: rgba(255,255,255,0.1);
74.         color: #fff;
75.         transform: translateX(4px);
76.     }
77.
78.     .sidebar a.tambah {
79.         background: var(--gold);
80.         color: var(--brown);
81.         font-weight: 600;
82.         margin-top: 12px;
83.         text-align: center;
84.     }
85.
86.     .sidebar a.tambah:hover {
87.         opacity: 0.9;
88.         transform: translateY(-2px);
89.     }
90.
91.     .main {
92.         flex: 1;
93.         padding: 48px 56px;
94.     }
95.
96.     .page-header {
97.         display: flex;
98.         align-items: center;
99.         justify-content: space-between;
100.        margin-bottom: 36px;
101.    }
102.
103.    .page-title {
104.        font-family: 'Playfair Display', serif;
105.        font-size: 2rem;
106.        color: var(--brown);
107.    }
108.
109.    .alert {
110.        padding: 14px 24px;
111.        border-radius: 12px;
112.        margin-bottom: 28px;
113.        font-weight: 500;
114.        font-size: 0.92rem;
115.    }
116.    .alert-success { background: #d1fae5; color: #065f46;
border-left: 4px solid #10b981; }
117.    .alert-error { background: #fee2e2; color: #991b1b;
border-left: 4px solid #ef4444; }
```

```
118.
119.     .table-wrap {
120.         background: #fff;
121.         border-radius: 18px;
122.         overflow: hidden;
123.         box-shadow: 0 8px 32px rgba(61,43,31,0.08);
124.     }
125.
126.     table {
127.         width: 100%;
128.         border-collapse: collapse;
129.     }
130.
131.     thead tr {
132.         background: var(--brown);
133.         color: #fff;
134.     }
135.
136.     thead th {
137.         padding: 18px 20px;
138.         text-align: left;
139.         font-size: 0.85rem;
140.         font-weight: 600;
141.         letter-spacing: 0.05em;
142.         text-transform: uppercase;
143.     }
144.
145.     tbody tr {
146.         border-bottom: 1px solid #f2ede4;
147.         transition: background 0.15s;
148.     }
149.
150.     tbody tr:hover { background: var(--light); }
151.
152.     tbody td {
153.         padding: 16px 20px;
154.         font-size: 0.92rem;
155.         color: #444;
156.         vertical-align: middle;
157.     }
158.
159.     .badge {
160.         display: inline-block;
161.         padding: 4px 12px;
162.         border-radius: 20px;
163.         font-size: 0.8rem;
164.         font-weight: 600;
165.         background: #ede8df;
166.         color: var(--brown);
167.     }
168.
169.     .actions {
```

```

170.         display: flex;
171.         gap: 8px;
172.         align-items: center;
173.     }
174.
175.     .btn {
176.         display: inline-block;
177.         padding: 8px 16px;
178.         border-radius: 8px;
179.         font-size: 0.85rem;
180.         font-weight: 600;
181.         text-decoration: none;
182.         border: none;
183.         cursor: pointer;
184.         transition: all 0.18s ease;
185.     }
186.
187.     .btn-hapus { background: var(--rust); color: #fff;
188. }
189.     .btn-hapus:hover { opacity: 0.9; transform:
190. translateY(-1px); }
191.     .btn-ubah { background: var(--gold); color: var(-
192. -brown); }
193.     .btn-ubah:hover { opacity: 0.9; transform:
194. translateY(-1px); }
195.
196.     .empty-state {
197.         text-align: center;
198.         padding: 56px;
199.         color: #aaa;
200.         font-size: 1rem;
201.     }
202. </style>
203. </head>
204. <body>
205. <aside class="sidebar">
206.     <h2>📖 Perpustakaan</h2>
207.     <a href="Index.php">🏠 Beranda</a>
208.     <a href="Member.php" class="active">👤 Member</a>
209.     <a href="Buku.php">📖 Buku</a>
210.     <a href="Peminjaman.php">🔑 Peminjaman</a>
211.     <a href="FormMember.php" class="tambah">+    Tambah
212. Member</a>
213. </aside>
214.
215. <main class="main">
216.     <div class="page-header">
217.         <h1 class="page-title">Daftar Member</h1>
218.     </div>

```

```

216.     <?php if ($pesan === 'hapus_berhasil'): ?>
217.         <div class="alert alert-success">✔ Data member
berhasil dihapus.</div>
218.     <?php elseif ($pesan === 'simpan_berhasil'): ?>
219.         <div class="alert alert-success">✔ Data member
berhasil disimpan.</div>
220.     <?php elseif ($pesan === 'ubah_berhasil'): ?>
221.         <div class="alert alert-success">✔ Data member
berhasil diperbarui.</div>
222.     <?php endif; ?>
223.
224.     <div class="table-wrap">
225.         <table>
226.             <thead>
227.                 <tr>
228.                     <th>ID</th>
229.                     <th>Nama Member</th>
230.                     <th>No. Member</th>
231.                     <th>Alamat</th>
232.                     <th>Tgl Mendaftar</th>
233.                     <th>Tgl Terakhir Bayar</th>
234.                     <th>Ops</th>
235.                 </tr>
236.             </thead>
237.             <tbody>
238.                 <?php if (empty($members)): ?>
239.                     <tr><td colspan="7" class="empty-
state">Belum ada data member.</td></tr>
240.                 <?php else: ?>
241.                     <?php foreach ($members as $m): ?>
242.                         <tr>
243.                             <td><span class="badge"><?=
htmlspecialchars($m['id_member']) ?></span></td>
244.                             <td><strong><?=
htmlspecialchars($m['nama_member']) ?></strong></td>
245.                             <td><?=
htmlspecialchars($m['nomor_member']) ?></td>
246.                             <td><?= htmlspecialchars($m['alamat']
?? '-' ) ?></td>
247.                             <td><?=
htmlspecialchars($m['tgl_mendaftar'] ?? '-' ) ?></td>
248.                             <td><?=
htmlspecialchars($m['tgl_terakhir_bayar'] ?? '-' ) ?></td>
249.                             <td style="display:flex;gap:6px;align-
items:center;">
250.                                 <a href="Member.php?hapus=<?=
$m['id_member'] ?>"
251.                                     class="btn btn-hapus"
252.                                     onclick="return confirm('Yakin
hapus member ini?')">Hapus</a>

```

253.	<code>\$m['id_member'] ?>"</code>	<code><a href="FormMember.php?id=<?="</code>
254.		<code>class="btn btn-ubah">Ubah</code>
255.		<code></td></code>
256.		<code></tr></code>
257.		<code><?php endforeach; ?></code>
258.		<code><?php endif; ?></code>
259.		<code></tbody></code>
260.		<code></table></code>
261.		<code></div></code>
262.	<code></main></code>	
263.		
264.	<code></body></code>	
265.	<code></html></code>	

Tabel 6. Source Code Buku.php

1.	<code><?php</code>
2.	<code>require_once 'Model.php';</code>
3.	
4.	<code>if (isset(\$_GET['hapus'])) {</code>
5.	<code> \$id = \$_GET['hapus'];</code>
6.	<code> deleteBuku(\$id);</code>
7.	<code> header("Location: Buku.php?pesan=hapus_berhasil");</code>
8.	<code> exit;</code>
9.	<code>}</code>
10.	
11.	<code>\$bukus = getAllBuku();</code>
12.	<code>\$pesan = \$_GET['pesan'] ?? '';</code>
13.	<code>?></code>
14.	<code><!DOCTYPE html></code>
15.	<code><html lang="id"></code>
16.	<code><head></code>
17.	<code> <meta charset="UTF-8"></code>
18.	<code> <meta name="viewport" content="width=device-width,</code>
	<code>initial-scale=1.0"></code>
19.	<code> <title>Data Buku - Perpustakaan</title></code>
20.	<code> <link</code>
	<code>href="https://fonts.googleapis.com/css2?family=Playfair+Dis</code>
	<code>play:wght@700&family=DM+Sans:wght@400;500;600&display=swap"</code>
	<code>rel="stylesheet"></code>
21.	<code> <style></code>
22.	<code> :root {</code>
23.	<code> --cream: #F5F0E8;</code>
24.	<code> --brown: #3D2B1F;</code>
25.	<code> --gold: #C8960C;</code>
26.	<code> --rust: #A63D2F;</code>
27.	<code> --sage: #4A6741;</code>
28.	<code> --light: #FAF8F4;</code>
29.	<code> }</code>
30.	<code> * { margin: 0; padding: 0; box-sizing: border-box; }</code>
31.	
32.	<code> body {</code>

```
33.         min-height: 100vh;
34.         background: var(--cream);
35.         font-family: 'DM Sans', sans-serif;
36.         display: flex;
37.     }
38.
39.     .sidebar {
40.         width: 15%;
41.         min-width: 250px;
42.         min-height: 100vh;
43.         background: var(--brown);
44.         padding: 40px 24px;
45.         display: flex;
46.         flex-direction: column;
47.         gap: 10px;
48.         flex-shrink: 0;
49.         box-shadow: 4px 0 24px rgba(61,43,31,0.05);
50.     }
51.
52.     .sidebar h2 {
53.         font-family: 'Playfair Display', serif;
54.         color: #fff;
55.         font-size: 1.3rem;
56.         margin-bottom: 8px;
57.         padding-bottom: 20px;
58.         border-bottom: 1px solid rgba(255,255,255,0.15);
59.     }
60.
61.     .sidebar a {
62.         display: block;
63.         padding: 12px 16px;
64.         color: rgba(255,255,255,0.75);
65.         text-decoration: none;
66.         border-radius: 10px;
67.         font-size: 0.95rem;
68.         font-weight: 500;
69.         transition: all 0.2s ease;
70.     }
71.
72.     .sidebar a:hover, .sidebar a.active {
73.         background: rgba(255,255,255,0.1);
74.         color: #fff;
75.         transform: translateX(4px);
76.     }
77.
78.     .sidebar a.tambah {
79.         background: var(--gold);
80.         color: var(--brown);
81.         font-weight: 600;
82.         margin-top: 12px;
83.         text-align: center;
84.     }
```

```
85.
86.     .sidebar a.tambah:hover {
87.         opacity: 0.9;
88.         transform: translateY(-2px);
89.     }
90.
91.     .main {
92.         flex: 1;
93.         padding: 48px 56px;
94.     }
95.
96.     .page-header {
97.         display: flex;
98.         align-items: center;
99.         justify-content: space-between;
100.        margin-bottom: 36px;
101.    }
102.
103.    .page-title {
104.        font-family: 'Playfair Display', serif;
105.        font-size: 2rem;
106.        color: var(--brown);
107.    }
108.
109.    .alert {
110.        padding: 14px 24px;
111.        border-radius: 12px;
112.        margin-bottom: 28px;
113.        font-weight: 500;
114.        font-size: 0.92rem;
115.    }
116.    .alert-success { background: #d1fae5; color:
117.    #065f46; border-left: 4px solid #10b981; }
118.    .alert-error { background: #fee2e2; color: #991b1b;
119.    border-left: 4px solid #ef4444; }
120.
121.    .table-wrap {
122.        background: #fff;
123.        border-radius: 18px;
124.        overflow: hidden;
125.        box-shadow: 0 8px 32px rgba(61,43,31,0.08);
126.    }
127.
128.    table {
129.        width: 100%;
130.        border-collapse: collapse;
131.    }
132.
133.    thead tr {
134.        background: var(--brown);
135.        color: #fff;
```



```
135.
136.     thead th {
137.         padding: 18px 20px;
138.         text-align: left;
139.         font-size: 0.85rem;
140.         font-weight: 600;
141.         letter-spacing: 0.05em;
142.         text-transform: uppercase;
143.     }
144.
145.     tbody tr {
146.         border-bottom: 1px solid #f2ede4;
147.         transition: background 0.15s;
148.     }
149.
150.     tbody tr:hover { background: var(--light); }
151.
152.     tbody td {
153.         padding: 16px 20px;
154.         font-size: 0.92rem;
155.         color: #444;
156.         vertical-align: middle;
157.     }
158.
159.     .badge {
160.         display: inline-block;
161.         padding: 4px 12px;
162.         border-radius: 20px;
163.         font-size: 0.8rem;
164.         font-weight: 600;
165.         background: #ede8df;
166.         color: var(--brown);
167.     }
168.
169.     .actions {
170.         display: flex;
171.         gap: 8px;
172.         align-items: center;
173.     }
174.
175.     .btn {
176.         display: inline-block;
177.         padding: 8px 16px;
178.         border-radius: 8px;
179.         font-size: 0.85rem;
180.         font-weight: 600;
181.         text-decoration: none;
182.         border: none;
183.         cursor: pointer;
184.         transition: all 0.18s ease;
185.     }
186.
```

```

187.         .btn-hapus { background: var(--rust); color: #fff;
188.     }
189.         .btn-hapus:hover { opacity: 0.9; transform:
190.     translateY(-1px); }
191.         .btn-ubah { background: var(--gold); color: var(-
192.     -brown); }
193.         .btn-ubah:hover { opacity: 0.9; transform:
194.     translateY(-1px); }
195.         .empty-state {
196.             text-align: center;
197.             padding: 56px;
198.             color: #aaa;
199.             font-size: 1rem;
200.         }
201.     </style>
202. </head>
203. <body>
204. <aside class="sidebar">
205.     <h2>📖 Perpustakaan</h2>
206.     <a href="Index.php">🏠 Beranda</a>
207.     <a href="Member.php">👤 Member</a>
208.     <a href="Buku.php" class="active">📖 Buku</a>
209.     <a href="Peminjaman.php">📌 Peminjaman</a>
210.     <a href="FormBuku.php" class="tambah">+ Tambah Buku</a>
211. </aside>
212. <main class="main">
213.     <div class="page-header">
214.         <h1 class="page-title">Daftar Buku</h1>
215.     </div>
216.     <?php if ($pesan === 'hapus_berhasil'): ?>
217.         <div class="alert alert-success">✅ Data buku
218.     berhasil dihapus.</div>
219.     <?php elseif ($pesan === 'simpan_berhasil'): ?>
220.         <div class="alert alert-success">✅ Data buku
221.     berhasil disimpan.</div>
222.     <?php elseif ($pesan === 'ubah_berhasil'): ?>
223.         <div class="alert alert-success">✅ Data buku
224.     berhasil diperbarui.</div>
225.     <?php endif; ?>
226.     <div class="table-wrap">
227.         <table>
228.             <thead>
229.                 <tr>
230.                     <th>ID</th>
231.                     <th>Judul Buku</th>

```

230.	<th>Penulis</th>
231.	<th>Penerbit</th>
232.	<th>Tahun Terbit</th>
233.	<th>Ops</th>
234.	</tr>
235.	</thead>
236.	<tbody>
237.	<?php if (empty(\$bukus)): ?>
238.	<tr><td colspan="6" class="empty-state">Belum ada data buku.</td></tr>
239.	<?php else: ?>
240.	<?php foreach (\$bukus as \$b): ?>
241.	<tr>
242.	<td><?=
243.	htmlspecialchars(\$b['id_buku']) ?></td>
244.	<td><?=
245.	htmlspecialchars(\$b['judul_buku']) ?></td>
246.	<td><?=
247.	htmlspecialchars(\$b['penulis']) ?></td>
248.	<td><?=
249.	htmlspecialchars(\$b['penerbit']) ?? '-' ?></td>
250.	<td><=
251.	htmlspecialchars(\$b['tahun_terbit']) ?? '-' ?></td>
252.	<td
253.	style="display:flex;gap:6px;align-items:center;">
254.	<a href="Buku.php?hapus=<?=
255.	\$b['id_buku'] ?>"
256.	class="btn btn-hapus"
257.	onclick="return
258.	confirm('Yakin hapus buku ini?')">Hapus
259.	<a href="FormBuku.php?id=<?=
260.	\$b['id_buku'] ?>"
261.	class="btn
262.	btn-
263.	ubah">Ubah
	</td>
	</tr>
	<?php endforeach; ?>
	<?php endif; ?>
	</tbody>
	</table>
	</div>
	</main>
	</body>
	</html>

Tabel 7. Source Code Peminjaman.php

1.	<?php
2.	require_once 'Model.php';
3.	
4.	if (isset(\$_GET['hapus'])) {

```

5.     $id = $_GET['hapus'];
6.     deletePeminjaman($id);
7.     header("Location:
Peminjaman.php?pesan=hapus_berhasil");
8.     exit;
9. }
10.
11. if (isset($_GET['selesai'])) {
12.     $id = $_GET['selesai'];
13.
14.     // Mengeksekusi fungsi baru Anda dan mengecek hasilnya
15.     $eksekusi = selesaikanPeminjaman($id);
16.
17.     if ($eksekusi) {
18.         header("Location:
Peminjaman.php?pesan=selesai_berhasil");
19.     } else {
20.         header("Location:
Peminjaman.php?pesan=selesai_gagal");
21.     }
22.     exit;
23. }
24.
25. $peminjamans = getAllPeminjaman();
26. $pesan       = $_GET['pesan'] ?? '';
27.
28. function hitungStatus($row) {
29.     if (($row['status'] ?? 'aktif') === 'selesai') return
'selesai';
30.     $today = date('Y-m-d');
31.     if ($today > $row['tgl_kembali']) return 'overdue';
32.     if ($today === $row['tgl_kembali']) return 'today';
33.     return 'aktif';
34. }
35. ?>
36. <!DOCTYPE html>
37. <html lang="id">
38. <head>
39.     <meta charset="UTF-8">
40.     <meta name="viewport" content="width=device-width,
initial-scale=1.0">
41.     <title>Data Peminjaman - Perpustakaan</title>
42.     <link
href="https://fonts.googleapis.com/css2?family=Playfair+Dis
play:wght@700&family=DM+Sans:wght@400;500;600&display=swap"
rel="stylesheet">
43.     <style>
44.         :root {
45.             --cream: #F5F0E8;
46.             --brown: #3D2B1F;
47.             --gold: #C8960C;
48.             --rust: #A63D2F;

```

```
49.         --sage: #4A6741;
50.         --blue: #1e4d8c;
51.         --light: #FAF8F4;
52.     }
53.     * { margin: 0; padding: 0; box-sizing: border-box; }
54.
55.     body {
56.         min-height: 100vh;
57.         background: var(--cream);
58.         font-family: 'DM Sans', sans-serif;
59.         display: flex;
60.     }
61.
62.     .sidebar {
63.         width: 15%;
64.         min-width: 250px;
65.         min-height: 100vh;
66.         background: var(--brown);
67.         padding: 40px 24px;
68.         display: flex;
69.         flex-direction: column;
70.         gap: 10px;
71.         flex-shrink: 0;
72.         box-shadow: 4px 0 24px rgba(61,43,31,0.05);
73.     }
74.
75.     .sidebar h2 {
76.         font-family: 'Playfair Display', serif;
77.         color: #fff;
78.         font-size: 1.3rem;
79.         margin-bottom: 8px;
80.         padding-bottom: 20px;
81.         border-bottom: 1px solid rgba(255,255,255,0.15);
82.     }
83.
84.     .sidebar a {
85.         display: block;
86.         padding: 12px 16px;
87.         color: rgba(255,255,255,0.75);
88.         text-decoration: none;
89.         border-radius: 10px;
90.         font-size: 0.95rem;
91.         font-weight: 500;
92.         transition: all 0.2s ease;
93.     }
94.
95.     .sidebar a:hover, .sidebar a.active {
96.         background: rgba(255,255,255,0.1);
97.         color: #fff;
98.         transform: translateX(4px);
99.     }
100.
```

```
101.     .sidebar a.tambah {
102.         background: var(--gold);
103.         color: var(--brown);
104.         font-weight: 600;
105.         margin-top: 12px;
106.         text-align: center;
107.     }
108.
109.     .sidebar a.tambah:hover {
110.         opacity: 0.9;
111.         transform: translateY(-2px);
112.     }
113.
114.     .main {
115.         flex: 1;
116.         padding: 48px 56px;
117.     }
118.
119.     .page-header {
120.         display: flex;
121.         align-items: center;
122.         justify-content: space-between;
123.         margin-bottom: 36px;
124.     }
125.
126.     .page-title {
127.         font-family: 'Playfair Display', serif;
128.         font-size: 2rem;
129.         color: var(--brown);
130.     }
131.
132.     .alert {
133.         padding: 14px 24px;
134.         border-radius: 12px;
135.         margin-bottom: 28px;
136.         font-weight: 500;
137.         font-size: 0.92rem;
138.     }
139.     .alert-success { background: #d1fae5; color:
#065f46; border-left: 4px solid #10b981; }
140.     .alert-error { background: #fee2e2; color: #991b1b;
border-left: 4px solid #ef4444; }
141.
142.     .table-wrap {
143.         background: #fff;
144.         border-radius: 18px;
145.         overflow: hidden;
146.         box-shadow: 0 8px 32px rgba(61,43,31,0.08);
147.     }
148.
149.     table {
150.         width: 100%;
```

```
151.         border-collapse: collapse;
152.     }
153.
154.     thead tr {
155.         background: var(--brown);
156.         color: #fff;
157.     }
158.
159.     thead th {
160.         padding: 18px 20px;
161.         text-align: left;
162.         font-size: 0.85rem;
163.         font-weight: 600;
164.         letter-spacing: 0.05em;
165.         text-transform: uppercase;
166.     }
167.
168.     tbody tr {
169.         border-bottom: 1px solid #f2ede4;
170.         transition: background 0.15s;
171.     }
172.
173.     tbody tr:hover { background: var(--light); }
174.
175.     tbody td {
176.         padding: 16px 20px;
177.         font-size: 0.92rem;
178.         color: #444;
179.         vertical-align: middle;
180.     }
181.
182.     .badge {
183.         display: inline-block;
184.         padding: 4px 12px;
185.         border-radius: 20px;
186.         font-size: 0.8rem;
187.         font-weight: 600;
188.         background: #ede8df;
189.         color: var(--brown);
190.     }
191.
192.     .actions {
193.         display: flex;
194.         gap: 8px;
195.         align-items: center;
196.     }
197.
198.     .btn {
199.         display: inline-block;
200.         padding: 8px 16px;
201.         border-radius: 8px;
202.         font-size: 0.85rem;
```

```

203.         font-weight: 600;
204.         text-decoration: none;
205.         border: none;
206.         cursor: pointer;
207.         transition: all 0.18s ease;
208.     }
209.
210.     .btn-hapus { background: var(--rust); color: #fff;
211. }
212.     .btn-hapus:hover { opacity: 0.9; transform:
translateY(-1px); }
213.     .btn-ubah { background: var(--gold); color: var(-
-brown); }
214.     .btn-ubah:hover { opacity: 0.9; transform:
translateY(-1px); }
215.     .btn-selesai { background: var(--sage); color: #fff;
216. }
217.     .btn-selesai:hover { background: #3a5432; }
218.     .btn-disabled { background: #e5e7eb; color: #9ca3af;
219. cursor: not-allowed; pointer-events: none; }
220.
221.     .empty-state {
222.         text-align: center;
223.         padding: 56px;
224.         color: #aaa;
225.         font-size: 1rem;
226.     }
227. }
228. </style>
229. </head>
230. <body>
231. <aside class="sidebar">
232.     <h2>📖 Perpustakaan</h2>
233.     <a href="Index.php">🏠 Beranda</a>
234.     <a href="Member.php">👤 Member</a>
235.     <a href="Buku.php">📖 Buku</a>
236.     <a href="Peminjaman.php" class="active">🔥
Peminjaman</a>
237.     <a href="FormPeminjaman.php" class="tambah">+
Tambah
Peminjaman</a>
238. </aside>
239.
240. <main class="main">
241.     <div class="page-header">
242.         <h1 class="page-title">Data Peminjaman</h1>
243.     </div>
244.
245.     <?php if ($pesan === 'hapus_berhasil'): ?>
246.         <div class="alert alert-success">✅ Data peminjaman
berhasil dihapus.</div>

```



```

244.     <?php elseif ($pesan === 'simpan_berhasil'): ?>
245.         <div class="alert alert-success">✔ Data peminjaman
berhasil disimpan.</div>
246.     <?php elseif ($pesan === 'ubah_berhasil'): ?>
247.         <div class="alert alert-success">✔ Data peminjaman
berhasil diperbarui.</div>
248.     <?php elseif ($pesan === 'selesai_berhasil'): ?>
249.         <div class="alert alert-success">✔ Peminjaman
berhasil diselesaikan.</div>
250.     <?php elseif ($pesan === 'selesai_gagal'): ?>
251.         <div class="alert alert-danger">✗ Gagal
menyelesaikan peminjaman atau data tidak ditemukan/sudah
selesai.</div>
252.     <?php endif; ?>
253.
254.     <div class="table-wrap">
255.         <table>
256.             <thead>
257.                 <tr>
258.                     <th>ID</th>
259.                     <th>Nama Member</th>
260.                     <th>No. Member</th>
261.                     <th>Judul Buku</th>
262.                     <th>Tgl Pinjam</th>
263.                     <th>Tgl Kembali</th>
264.                     <th>Status</th>
265.                     <th>Opsisi</th>
266.                 </tr>
267.             </thead>
268.             <tbody>
269.                 <?php if (empty($peminjamans)): ?>
270.                     <tr><td colspan="8" class="empty-
state">Belum ada data peminjaman.</td></tr>
271.                 <?php else: ?>
272.                     <?php foreach ($peminjamans as $p): ?>
273.                         <?php $status = hitungStatus($p); ?>
274.                         <tr class="<?= $status === 'selesai' ?
'row-selesai' : '' ?>">
275.                             <td><span class="badge badge-id"><?=
htmlspecialchars($p['id_peminjaman']) ?></span></td>
276.                             <td><strong><?=
htmlspecialchars($p['nama_member']) ?></strong></td>
277.                             <td><?=
htmlspecialchars($p['nomor_member']) ?></td>
278.                             <td><?=
htmlspecialchars($p['judul_buku']) ?></td>
279.                             <td><?=
htmlspecialchars($p['tgl_pinjam']) ?></td>
280.                             <td><?=
htmlspecialchars($p['tgl_kembali']) ?></td>
281.                             <td>

```

```

282.                                     <?php      if      ($status      ===
'selesai'): ?>
283.                                     <span   class="badge   badge-
selesai">✓ Selesai</span>
284.                                     <?php      elseif      ($status      ===
'overdue'): ?>
285.                                     <span   class="badge   badge-
overdue">⚠ Terlambat</span>
286.                                     <?php      elseif      ($status      ===
'today'): ?>
287.                                     <span   class="badge   badge-
today">🕒 Hari Ini</span>
288.                                     <?php else: ?>
289.                                     <span   class="badge   badge-
aktif">✓ Aktif</span>
290.                                     <?php endif; ?>
291.                                     </td>
292.                                     <td>
293.                                     <div class="actions">
294.                                     <?php      if      ($status      !==
'selesai'): ?>
295.                                     <a
href="Peminjaman.php?selesai=<?= $p['id_peminjaman'] ?>"
296.                                     class="btn      btn-
selesai"
297.                                     onclick="return
confirm('Tandai      peminjaman      ini      sebagai
selesai?')">Selesai</a>
298.                                     <a
href="FormPeminjaman.php?id=<?= $p['id_peminjaman'] ?>"
299.                                     class="btn      btn-
ubah">Ubah</a>
300.                                     <?php else: ?>
301.                                     <span   class="btn   btn-
disabled">Selesai</span>
302.                                     <?php endif; ?>
303.                                     <a
href="Peminjaman.php?hapus=<?= $p['id_peminjaman'] ?>"
304.                                     class="btn btn-hapus"
305.                                     onclick="return
confirm('Yakin hapus data peminjaman ini?')">Hapus</a>
306.                                     </div>
307.                                     </td>
308.                                     </tr>
309.                                     <?php endforeach; ?>
310.                                     <?php endif; ?>
311.                                 </tbody>
312.                            </table>
313.                    </div>
314.    </main>
315.

```

316.	</body>
317.	</html>

Tabel 8. Source Code FormMember.php

1.	<?php
2.	require_once 'Model.php';
3.	
4.	\$id = \$_GET['id'] ?? null;
5.	\$data = null;
6.	\$isEdit = false;
7.	\$errors = [];
8.	
9.	if (\$id) {
10.	\$data = getMemberById(\$id);
11.	\$isEdit = true;
12.	}
13.	
14.	if (\$_SERVER['REQUEST_METHOD'] === 'POST') {
15.	\$nama_member = trim(\$_POST['nama_member'] ?? '');
16.	\$nomor_member = trim(\$_POST['nomor_member'] ?? '');
17.	\$alamat = trim(\$_POST['alamat'] ?? '');
18.	\$ttl_mendaftar = trim(\$_POST['ttl_mendaftar'] ?? '');
19.	\$ttl_terakhir_bayar = trim(\$_POST['ttl_terakhir_bayar'] ?? '');
20.	\$id_post = \$_POST['id_member'] ?? null;
21.	
22.	if (empty(\$nama_member)) \$errors[] = 'Nama member wajib diisi.';
23.	if (empty(\$nomor_member)) \$errors[] = 'Nomor member wajib diisi.';
24.	if (empty(\$ttl_mendaftar)) \$errors[] = 'Tanggal mendaftar wajib diisi.';
25.	
26.	if (empty(\$errors)) {
27.	if (\$id_post) {
28.	updateMember(\$id_post, \$nama_member, \$nomor_member, \$alamat, \$ttl_mendaftar, \$ttl_terakhir_bayar);
29.	header("Location: Member.php?pesan=ubah_berhasil");
30.	} else {
31.	insertMember(\$nama_member, \$nomor_member, \$alamat, \$ttl_mendaftar, \$ttl_terakhir_bayar);
32.	header("Location: Member.php?pesan=simpan_berhasil");
33.	}
34.	exit;
35.	}
36.	
37.	\$data = [
38.	'id_member' => \$id_post,

```

39.         'nama_member'          => $nama_member,
40.         'nomor_member'         => $nomor_member,
41.         'alamat'               => $alamat,
42.         'tgl_mendaftar'        => $tgl_mendaftar,
43.         'tgl_terkahir_bayar'   => $tgl_terkahir_bayar,
44.     ];
45.     $isEdit = !empty($id_post);
46. }
47.
48. $judulForm = $isEdit ? 'Edit Data Member' : 'Tambah Data
Member';
49. $namaBtn    = $isEdit ? 'Simpan Perubahan' : 'Daftarkan
Member';
50. ?>
51. <!DOCTYPE html>
52. <html lang="id">
53. <head>
54.     <meta charset="UTF-8">
55.     <meta name="viewport" content="width=device-width,
initial-scale=1.0">
56.     <title><?= $judulForm ?> - Perpustakaan</title>
57.     <link
href="https://fonts.googleapis.com/css2?family=Playfair+Dis
play:wght@700&family=DM+Sans:wght@400;500;600&display=swap"
rel="stylesheet">
58.     <style>
59.         :root {
60.             --cream: #F5F0E8;
61.             --brown: #3D2B1F;
62.             --gold: #C8960C;
63.             --rust: #A63D2F;
64.             --sage: #4A6741;
65.         }
66.         * { margin: 0; padding: 0; box-sizing: border-box; }
67.
68.         body {
69.             min-height: 100vh;
70.             background: var(--cream);
71.             font-family: 'DM Sans', sans-serif;
72.         }
73.
74.         nav {
75.             background: var(--brown);
76.             min-height: 7vh;
77.             padding: 0 48px;
78.             display: flex;
79.             align-items: center;
80.             justify-content: space-between;
81.             box-shadow: 0 4px 16px rgba(61,43,31,0.12);
82.         }
83.
84.         nav .brand {

```

```
85.         font-family: 'Playfair Display', serif;
86.         color: #fff;
87.         font-size: 1.3rem;
88.         font-weight: 700;
89.     }
90.
91.     nav a {
92.         color: rgba(255,255,255,0.85);
93.         text-decoration: none;
94.         font-size: 0.95rem;
95.         padding: 10px 20px;
96.         border: 1.5px solid rgba(255,255,255,0.35);
97.         border-radius: 10px;
98.         font-weight: 500;
99.         transition: all 0.2s ease;
100.    }
101.
102.    nav a:hover {
103.        background: rgba(255,255,255,0.1);
104.        color: #fff;
105.        border-color: #fff;
106.    }
107.
108.    .container {
109.        flex: 1;
110.        width: 100%;
111.        display: flex;
112.        align-items: center;
113.        justify-content: center;
114.        padding: 40px 24px;
115.    }
116.
117.    .form-card {
118.        background: #fff;
119.        border-radius: 24px;
120.        padding: 56px;
121.        box-shadow: 0 12px 48px rgba(61,43,31,0.12);
122.        border-top: 6px solid var(--sage);
123.        width: 60%;
124.        max-width: 750px;
125.        min-width: 340px;
126.    }
127.
128.    .form-card h1 {
129.        font-family: 'Playfair Display', serif;
130.        font-size: 1.9rem;
131.        color: var(--brown);
132.        margin-bottom: 8px;
133.    }
134.
135.    .form-card p.sub {
136.        color: #999;
```

```

137.         font-size: 0.88rem;
138.         margin-bottom: 32px;
139.     }
140.
141.     .alert-errors {
142.         background: #fee2e2;
143.         border-left: 4px solid #ef4444;
144.         border-radius: 8px;
145.         padding: 14px 18px;
146.         margin-bottom: 24px;
147.         font-size: 0.88rem;
148.         color: #991b1b;
149.     }
150.
151.     .alert-errors ul { padding-left: 18px; }
152.
153.     .field {
154.         margin-bottom: 22px;
155.     }
156.
157.     label {
158.         display: block;
159.         font-weight: 600;
160.         font-size: 0.88rem;
161.         color: var(--brown);
162.         margin-bottom: 7px;
163.     }
164.
165.     label span.req { color: var(--rust); }
166.
167.     input[type="text"],
168.     input[type="date"],
169.     input[type="datetime-local"],
170.     textarea {
171.         width: 100%;
172.         padding: 12px 16px;
173.         border: 1.5px solid #ddd;
174.         border-radius: 10px;
175.         font-family: 'DM Sans', sans-serif;
176.         font-size: 0.95rem;
177.         color: #333;
178.         transition: border-color 0.18s;
179.         outline: none;
180.     }
181.
182.     input:focus, textarea:focus {
183.         border-color: var(--sage);
184.     }
185.
186.     textarea { resize: vertical; min-height: 90px; }
187.
188.     .btn-submit {

```

```

189.         width: 100%;
190.         padding: 14px;
191.         background: var(--sage);
192.         color: #fff;
193.         border: none;
194.         border-radius: 12px;
195.         font-family: 'DM Sans', sans-serif;
196.         font-size: 1rem;
197.         font-weight: 600;
198.         cursor: pointer;
199.         transition: all 0.2s;
200.         margin-top: 8px;
201.     }
202.
203.     .btn-submit:hover {
204.         background: #3a5432;
205.         transform: translateY(-2px);
206.         box-shadow: 0 8px 24px rgba(74,103,65,0.25);
207.     }
208. </style>
209. </head>
210. <body>
211.
212. <nav>
213.     <span class="brand">  Perpustakaan Digital</span>
214.     <a href="Member.php">← Kembali</a>
215. </nav>
216.
217. <div class="container">
218.     <div class="form-card">
219.         <h1><?= $judulForm ?></h1>
220.         <p class="sub">Lengkapi semua field yang wajib diisi
221.         (<span style="color:var(--rust)">*</span>)</p>
222.
223.         <?php if (!empty($errors)): ?>
224.             <div class="alert-errors">
225.                 <ul>
226.                     <?php foreach ($errors as $e): ?>
227.                         <li><?= htmlspecialchars($e) ?></li>
228.                     <?php endforeach; ?>
229.                 </ul>
230.             </div>
231.         <?php endif; ?>
232.
233.         <form method="POST" action="FormMember.php<?=
234. $isEdit ? 'id=' . ($data['id_member'] ?? '') : '' ?>">
235.             <?php if ($isEdit && isset($data['id_member'])):
236.                 <input type="hidden" name="id_member"
237. value="<?= htmlspecialchars($data['id_member']) ?>">
238.             <?php endif; ?>

```

```

236.
237.         <div class="field">
238.             <label>Nama                      Member          <span
class="req">*</span></label>
239.                 <input type="text" name="nama_member"
240.                     value="<?=
htmlspecialchars($data['nama_member'] ?? '') ?>"
241.                     placeholder="Masukkan nama lengkap">
242.             </div>
243.
244.         <div class="field">
245.             <label>Nomor                      Member          <span
class="req">*</span></label>
246.                 <input type="text" name="nomor_member"
247.                     value="<?=
htmlspecialchars($data['nomor_member'] ?? '') ?>"
248.                     maxlength="15"
249.                     placeholder="Contoh: M001">
250.             </div>
251.
252.         <div class="field">
253.             <label>Alamat</label>
254.             <textarea                                name="alamat"
placeholder="Masukkan alamat lengkap"><?=
htmlspecialchars($data['alamat'] ?? '') ?></textarea>
255.             </div>
256.
257.         <div class="field">
258.             <label>Tanggal                      Mendaftar    <span
class="req">*</span></label>
259.                 <input                                type="datetime-local"
name="tgl_mendaftar"
260.                     value="<?= htmlspecialchars(
261.                         isset($data['tgl_mendaftar']) &&
$data['tgl_mendaftar']
262.                             ?           date('Y-m-d\TH:i',
strtotime($data['tgl_mendaftar']))
263.                             : date('Y-m-d\TH:i')
264.                         ) ?>">
265.             </div>
266.
267.         <div class="field">
268.             <label>Tanggal Terakhir Bayar</label>
269.             <input type="date" name="tgl_terakhir_bayar"
270.                 value="<?=
htmlspecialchars($data['tgl_terakhir_bayar'] ?? '') ?>">
271.             </div>
272.
273.         <button type="submit" class="btn-submit"><?=
$namaBtn ?></button>
274.     </form>
275. </div>

```


276.	</div>
277.	
278.	</body>
279.	</html>

Tabel 9. Source Code FormBuku.php

1.	<?php
2.	require_once 'Model.php';
3.	
4.	\$id = \$_GET['id'] ?? null;
5.	\$data = null;
6.	\$isEdit = false;
7.	\$errors = [];
8.	
9.	if (\$id) {
10.	\$data = getBukuById(\$id);
11.	\$isEdit = true;
12.	}
13.	
14.	if (\$_SERVER['REQUEST_METHOD'] === 'POST') {
15.	\$judul_buku = trim(\$_POST['judul_buku']) ?? '';
16.	\$penulis = trim(\$_POST['penulis']) ?? '';
17.	\$penerbit = trim(\$_POST['penerbit']) ?? '';
18.	\$tahun_terbit = trim(\$_POST['tahun_terbit']) ?? '';
19.	\$id_post = \$_POST['id_buku'] ?? null;
20.	
21.	if (empty(\$judul_buku)) \$errors[] = 'Judul buku wajib diisi.';
22.	if (empty(\$penulis)) \$errors[] = 'Nama penulis wajib diisi.';
23.	if (!empty(\$tahun_terbit) && (!is_numeric(\$tahun_terbit) \$tahun_terbit < 1000 \$tahun_terbit > date('Y'))) {
24.	\$errors[] = 'Tahun terbit tidak valid.';
25.	}
26.	
27.	if (empty(\$errors)) {
28.	if (\$id_post) {
29.	updateBuku(\$id_post, \$judul_buku, \$penulis, \$penerbit, \$tahun_terbit);
30.	header("Location: Buku.php?pesan=ubah_berhasil");
31.	} else {
32.	insertBuku(\$judul_buku, \$penulis, \$penerbit, \$tahun_terbit);
33.	header("Location: Buku.php?pesan=simpan_berhasil");
34.	}
35.	exit;
36.	}
37.	
38.	\$data = [
39.	'id_buku' => \$id_post,

```

40.         'judul_buku'    => $judul_buku,
41.         'penulis'       => $penulis,
42.         'penerbit'      => $penerbit,
43.         'tahun_terbit'  => $tahun_terbit,
44.     ];
45.     $isEdit = !empty($id_post);
46. }
47.
48. $judulForm = $isEdit ? 'Edit Data Buku' : 'Tambah Data Buku';
49. $namaBtn    = $isEdit ? 'Simpan Perubahan' : 'Tambahkan Buku';
50. ?>
51. <!DOCTYPE html>
52. <html lang="id">
53. <head>
54.     <meta charset="UTF-8">
55.     <meta name="viewport" content="width=device-width,
initial-scale=1.0">
56.     <title><?= $judulForm ?> - Perpustakaan</title>
57.     <link
href="https://fonts.googleapis.com/css2?family=Playfair+Disp
lay:wght@700&family=DM+Sans:wght@400;500;600&display=swap"
rel="stylesheet">
58.     <style>
59.         :root {
60.             --cream: #F5F0E8;
61.             --brown: #3D2B1F;
62.             --gold: #C8960C;
63.             --rust: #A63D2F;
64.             --sage: #4A6741;
65.         }
66.         * { margin: 0; padding: 0; box-sizing: border-box; }
67.
68.         body {
69.             min-height: 100vh;
70.             background: var(--cream);
71.             font-family: 'DM Sans', sans-serif;
72.             display: flex;
73.             flex-direction: column;
74.         }
75.
76.         nav {
77.             background: var(--brown);
78.             min-height: 7vh;
79.             padding: 0 48px;
80.             display: flex;
81.             align-items: center;
82.             justify-content: space-between;
83.             box-shadow: 0 4px 16px rgba(61,43,31,0.12);
84.         }
85.
86.         nav .brand {
87.             font-family: 'Playfair Display', serif;

```

```
88.         color: #fff;
89.         font-size: 1.3rem;
90.         font-weight: 700;
91.     }
92.
93.     nav a {
94.         color: rgba(255,255,255,0.85);
95.         text-decoration: none;
96.         font-size: 0.95rem;
97.         padding: 10px 20px;
98.         border: 1.5px solid rgba(255,255,255,0.35);
99.         border-radius: 10px;
100.        font-weight: 500;
101.        transition: all 0.2s ease;
102.    }
103.
104.    nav a:hover {
105.        background: rgba(255,255,255,0.1);
106.        color: #fff;
107.        border-color: #fff;
108.    }
109.
110.    .container {
111.        flex: 1;
112.        width: 100%;
113.        display: flex;
114.        align-items: center;
115.        justify-content: center;
116.        padding: 40px 24px;
117.    }
118.
119.    .form-card {
120.        background: #fff;
121.        border-radius: 24px;
122.        padding: 56px;
123.        box-shadow: 0 12px 48px rgba(61,43,31,0.12);
124.        border-top: 6px solid var(--sage);
125.        width: 60%;
126.        max-width: 750px;
127.        min-width: 340px;
128.    }
129.
130.    .form-card h1 {
131.        font-family: 'Playfair Display', serif;
132.        font-size: 1.9rem;
133.        color: var(--brown);
134.        margin-bottom: 8px;
135.    }
136.
137.    .form-card p.sub {
138.        color: #999;
139.        font-size: 0.92rem;
```

```
140.         margin-bottom: 36px;
141.     }
142.
143.     .alert-errors {
144.         background: #fee2e2;
145.         border-left: 4px solid #ef4444;
146.         border-radius: 10px;
147.         padding: 16px 20px;
148.         margin-bottom: 28px;
149.         font-size: 0.9rem;
150.         color: #991b1b;
151.     }
152.
153.     .alert-errors ul { padding-left: 20px; }
154.
155.     .field { margin-bottom: 24px; }
156.
157.     label {
158.         display: block;
159.         font-weight: 600;
160.         font-size: 0.92rem;
161.         color: var(--brown);
162.         margin-bottom: 8px;
163.     }
164.
165.     label span.req { color: var(--rust); }
166.
167.     input[type="text"],
168.     input[type="number"] {
169.         width: 100%;
170.         padding: 14px 18px;
171.         border: 1.5px solid #dddad4;
172.         border-radius: 12px;
173.         font-family: 'DM Sans', sans-serif;
174.         font-size: 0.98rem;
175.         color: #333;
176.         transition: all 0.18s ease;
177.         outline: none;
178.     }
179.
180.     input:focus {
181.         border-color: var(--sage);
182.         box-shadow: 0 0 0 3px rgba(74,103,65,0.1);
183.     }
184.
185.     .btn-submit {
186.         width: 100%;
187.         padding: 16px;
188.         background: var(--sage);
189.         color: #fff;
190.         border: none;
191.         border-radius: 14px;
```

```

192.         font-family: 'DM Sans', sans-serif;
193.         font-size: 1.05rem;
194.         font-weight: 600;
195.         cursor: pointer;
196.         transition: all 0.22s ease;
197.         margin-top: 12px;
198.     }
199.
200.     .btn-submit:hover {
201.         background: #3a5432;
202.         transform: translateY(-2px);
203.         box-shadow: 0 8px 24px rgba(74,103,65,0.25);
204.     }
205. </style>
206. </head>
207. <body>
208.
209. <nav>
210.     <span class="brand">📖 Perpustakaan Digital</span>
211.     <a href="Buku.php">← Kembali</a>
212. </nav>
213.
214. <div class="container">
215.     <div class="form-card">
216.         <h1><?= $judulForm ?></h1>
217.         <p class="sub">Isi informasi buku dengan lengkap dan
benar.</p>
218.
219.         <?php if (!empty($errors)): ?>
220.             <div class="alert-errors">
221.                 <ul>
222.                     <?php foreach ($errors as $e): ?>
223.                         <li><?= htmlspecialchars($e) ?></li>
224.                     <?php endforeach; ?>
225.                 </ul>
226.             </div>
227.         <?php endif; ?>
228.
229.         <form method="POST" action="FormBuku.php<?= $isEdit
? '?'id=' . ($data['id_buku'] ?? '') : '' ?>">
230.             <?php if ($isEdit && isset($data['id_buku'])): ?>
231.                 <input type="hidden" name="id_buku"
value="<?= htmlspecialchars($data['id_buku']) ?>">
232.             <?php endif; ?>
233.
234.             <div class="field">
235.                 <label>Judul Buku <span
class="req">*</span></label>
236.                 <input type="text" name="judul_buku"
value="<?=
237.                 htmlspecialchars($data['judul_buku'] ?? '') ?>"

```

238.	placeholder="Masukkan judul buku">
239.	</div>
240.	
241.	<div class="field">
242.	<label>Penulis *</label>
243.	<input type="text" name="penulis"
244.	value="<?= htmlspecialchars(\$data['penulis'] ?? '') ?>"
245.	placeholder="Nama penulis">
246.	</div>
247.	
248.	<div class="field">
249.	<label>Penerbit</label>
250.	<input type="text" name="penerbit"
251.	value="<?= htmlspecialchars(\$data['penerbit'] ?? '') ?>"
252.	placeholder="Nama penerbit">
253.	</div>
254.	
255.	<div class="field">
256.	<label>Tahun Terbit</label>
257.	<input type="number" name="tahun_terbit"
258.	value="<?= htmlspecialchars(\$data['tahun_terbit'] ?? '') ?>"
259.	min="1000" max="<?= date('Y') ?>"
260.	placeholder="Contoh: 2023">
261.	</div>
262.	
263.	<button type="submit" class="btn-submit"><?= \$namaBtn ?></button>
264.	</form>
265.	</div>
266.	</div>
267.	
268.	</body>
269.	</html>

Tabel 10. Source Code FormPeminjaman.php

1.	<?php
2.	require_once 'Model.php';
3.	
4.	\$id = \$_GET['id'] ?? null;
5.	\$data = null;
6.	\$isEdit = false;
7.	\$errors = [];
8.	
9.	\$members = getAllMember();
10.	\$bukus = getAllBuku();
11.	
12.	if (\$id) {
13.	\$data = getPeminjamanById(\$id);

```

14.     $isEdit = true;
15. }
16.
17. if ($_SERVER['REQUEST_METHOD'] === 'POST') {
18.     $id_member = trim($_POST['id_member'] ?? '');
19.     $id_buku = trim($_POST['id_buku'] ?? '');
20.     $tgl_pinjam = trim($_POST['tgl_pinjam'] ?? '');
21.     $tgl_kembali = trim($_POST['tgl_kembali'] ?? '');
22.     $id_post = $_POST['id_peminjaman'] ?? null;
23.
24.     if (empty($id_member)) $errors[] = 'Member wajib
dipilih.';
25.     if (empty($id_buku)) $errors[] = 'Buku wajib
dipilih.';
26.     if (empty($tgl_pinjam)) $errors[] = 'Tanggal pinjam
wajib diisi.';
27.     if (empty($tgl_kembali)) $errors[] = 'Tanggal kembali
wajib diisi.';
28.
29.     if (!empty($tgl_pinjam) && !empty($tgl_kembali)) {
30.         if (strtotime($tgl_kembali) <
strtotime($tgl_pinjam)) {
31.             $errors[] = 'Tanggal kembali tidak boleh lebih
awal dari tanggal pinjam.';
32.         }
33.     }
34.
35.     if (empty($errors)) {
36.         if ($id_post) {
37.             updatePeminjaman($id_post, $id_member, $id_buku,
$tgl_pinjam, $tgl_kembali);
38.             header("Location:
Peminjaman.php?pesan=ubah_berhasil");
39.         } else {
40.             insertPeminjaman($id_member, $id_buku,
$tgl_pinjam, $tgl_kembali);
41.             header("Location:
Peminjaman.php?pesan=simpan_berhasil");
42.         }
43.         exit;
44.     }
45.
46.     $data = [
47.         'id_peminjaman' => $id_post,
48.         'id_member' => $id_member,
49.         'id_buku' => $id_buku,
50.         'tgl_pinjam' => $tgl_pinjam,
51.         'tgl_kembali' => $tgl_kembali,
52.     ];
53.     $isEdit = !empty($id_post);
54. }
55.

```

```

56. $judulForm = $sisEdit ? 'Edit Data Peminjaman' : 'Tambah Data
Peminjaman';
57. $namaBtn      = $sisEdit ? 'Simpan Perubahan' : 'Catat
Peminjaman';
58. ?>
59. <!DOCTYPE html>
60. <html lang="id">
61. <head>
62.     <meta charset="UTF-8">
63.     <meta name="viewport" content="width=device-width,
initial-scale=1.0">
64.     <title><?= $judulForm ?> - Perpustakaan</title>
65.     <link
href="https://fonts.googleapis.com/css2?family=Playfair+Disp
lay:wght@700&family=DM+Sans:wght@400;500;600&display=swap"
rel="stylesheet">
66.     <style>
67.         :root {
68.             --cream: #F5F0E8;
69.             --brown: #3D2B1F;
70.             --gold: #C8960C;
71.             --rust: #A63D2F;
72.             --sage: #4A6741;
73.         }
74.         * { margin: 0; padding: 0; box-sizing: border-box; }
75.
76.         body {
77.             min-height: 100vh;
78.             background: var(--cream);
79.             font-family: 'DM Sans', sans-serif;
80.         }
81.
82.         nav {
83.             background: var(--brown);
84.             min-height: 7vh;
85.             padding: 0 48px;
86.             display: flex;
87.             align-items: center;
88.             justify-content: space-between;
89.             box-shadow: 0 4px 16px rgba(61,43,31,0.12);
90.         }
91.
92.         nav .brand {
93.             font-family: 'Playfair Display', serif;
94.             color: #fff;
95.             font-size: 1.3rem;
96.             font-weight: 700;
97.         }
98.
99.         nav a {
100.             color: rgba(255,255,255,0.85);
101.             text-decoration: none;

```



```
102.         font-size: 0.95rem;
103.         padding: 10px 20px;
104.         border: 1.5px solid rgba(255,255,255,0.35);
105.         border-radius: 10px;
106.         font-weight: 500;
107.         transition: all 0.2s ease;
108.     }
109.
110.     nav a:hover {
111.         background: rgba(255,255,255,0.1);
112.         color: #fff;
113.         border-color: #fff;
114.     }
115.
116.     .container {
117.         flex: 1;
118.         width: 100%;
119.         display: flex;
120.         align-items: center;
121.         justify-content: center;
122.         padding: 40px 24px;
123.     }
124.
125.     .form-card {
126.         background: #fff;
127.         border-radius: 24px;
128.         padding: 56px;
129.         box-shadow: 0 12px 48px rgba(61,43,31,0.12);
130.         border-top: 6px solid var(--sage);
131.         width: 60%;
132.         max-width: 750px;
133.         min-width: 340px;
134.     }
135.
136.     .form-card h1 {
137.         font-family: 'Playfair Display', serif;
138.         font-size: 1.9rem;
139.         color: var(--brown);
140.         margin-bottom: 8px;
141.     }
142.
143.     .form-card p.sub {
144.         color: #777;
145.         font-size: 0.9rem;
146.         margin-bottom: 32px;
147.     }
148.
149.     .alert-errors {
150.         background: #fee2e2;
151.         border-left: 4px solid #ef4444;
152.         border-radius: 8px;
153.         padding: 14px 18px;
```

```
154.         margin-bottom: 24px;
155.         font-size: 0.88rem;
156.         color: #991b1b;
157.     }
158.
159.     .alert-errors ul { padding-left: 18px; }
160.
161.     .field { margin-bottom: 22px; }
162.
163.     label {
164.         display: block;
165.         font-weight: 600;
166.         font-size: 0.88rem;
167.         color: var(--brown);
168.         margin-bottom: 7px;
169.     }
170.
171.     label span.req { color: var(--rust); }
172.
173.     input[type="date"],
174.     select {
175.         width: 100%;
176.         padding: 12px 16px;
177.         border: 1.5px solid #ddd;
178.         border-radius: 10px;
179.         font-family: 'DM Sans', sans-serif;
180.         font-size: 0.95rem;
181.         color: #333;
182.         transition: border-color 0.18s;
183.         outline: none;
184.         background: #fff;
185.     }
186.
187.     input:focus, select:focus { border-color: var(--
sage); }
188.
189.     .row-2 {
190.         display: grid;
191.         grid-template-columns: 1fr 1fr;
192.         gap: 18px;
193.     }
194.
195.     input:disabled {
196.         background: #f5f5f5;
197.         color: #aaa;
198.         cursor: not-allowed;
199.     }
200.
201.     .hint {
202.         font-size: 0.78rem;
203.         color: #999;
204.         margin-top: 5px;
```

```

205.     }
206.
207.     .btn-submit {
208.         width: 100%;
209.         padding: 14px;
210.         background: var(--sage);
211.         color: #fff;
212.         border: none;
213.         border-radius: 12px;
214.         font-family: 'DM Sans', sans-serif;
215.         font-size: 1rem;
216.         font-weight: 600;
217.         cursor: pointer;
218.         transition: all 0.2s;
219.         margin-top: 8px;
220.     }
221.
222.     .btn-submit:hover {
223.         background: #3a5432;
224.         transform: translateY(-2px);
225.         box-shadow: 0 8px 20px rgba(74,103,65,0.25);
226.     }
227.
228.     input.error, select.error {
229.         border-color: #ef4444;
230.     }
231.
232.     /* Responsive Fallback untuk device mobile/tablet */
233.     @media (max-width: 768px) {
234.         .container {
235.             width: 92%;
236.             min-width: unset;
237.             margin: 32px auto;
238.         }
239.         .form-card {
240.             padding: 32px 24px;
241.         }
242.         .row-2 {
243.             grid-template-columns: 1fr;
244.             gap: 0px;
245.         }
246.     }
247. </style>
248. </head>
249. <body>
250.
251. <nav>
252.     <span class="brand"> Perpustakaan Digital</span>
253.     <a href="Peminjaman.php">← Kembali </a>
254. </nav>
255.

```

```

256. <div class="container">
257.     <div class="form-card">
258.         <h1><?= $judulForm ?></h1>
259.         <p class="sub">Pilih member dan buku, kemudian
tentukan tanggal peminjaman.</p>
260.
261.         <?php if (!empty($errors)): ?>
262.             <div class="alert-errors">
263.                 <ul>
264.                     <?php foreach ($errors as $e): ?>
265.                         <li><?= htmlspecialchars($e) ?></li>
266.                     <?php endforeach; ?>
267.                 </ul>
268.             </div>
269.         <?php endif; ?>
270.
271.         <form method="POST"
272.             action="FormPeminjaman.php<?= $isEdit ? 'id='
. ($data['id_peminjaman'] ?? '') : '' ?>"
273.             id="formPeminjaman">
274.
275.             <?php if ($isEdit &&
isset($data['id_peminjaman'])): ?>
276.                 <input type="hidden" name="id_peminjaman"
value="<?= htmlspecialchars($data['id_peminjaman']) ?>"
277.                 <?php endif; ?>
278.
279.                 <div class="field">
280.                     <label>Nama Member <span
class="req">*</span></label>
281.                     <select name="id_member" id="id_member"
required>
282.                         <option value="">-- Pilih Member --
</option>
283.                         <?php foreach ($members as $m): ?>
284.                             <option value="<?= $m['id_member']
?>"
285.                                 <?= (isset($data['id_member'])
&& $data['id_member'] == $m['id_member']) ? 'selected' : ''
?>>
286.                             <?=
htmlspecialchars($m['nama_member']) ?> (<?=
htmlspecialchars($m['nomor_member']) ?>)
287.                             </option>
288.                         <?php endforeach; ?>
289.                     </select>
290.                 </div>
291.
292.                 <div class="field">
293.                     <label>Judul Buku <span
class="req">*</span></label>

```

```

294.         <select      name="id_buku"      id="id_buku"
required>
295.             <option  value="">--    Pilih    Buku    --
</option>
296.             <?php foreach ($bukus as $b): ?>
297.                 <option value="<?= $b['id_buku'] ?>"
298.                     <?= (isset($data['id_buku']) &&
$data['id_buku'] == $b['id_buku']) ? 'selected' : '' ?>>
299.                     <?=
htmlspecialchars($b['judul_buku'])      ?>      -      <?=
htmlspecialchars($b['penulis']) ?>
300.                 </option>
301.             <?php endforeach; ?>
302.         </select>
303.     </div>
304.
305.         <div class="row-2">
306.             <div class="field">
307.                 <label>Tanggal          Pinjam          <span
class="req">*</span></label>
308.                 <input type="date"
309.                     name="tgl_pinjam"
310.                     id="tgl_pinjam"
311.                     value="<?=
htmlspecialchars($data['tgl_pinjam'] ?? date('Y-m-d')) ?>"
312.                     required>
313.             </div>
314.
315.             <div class="field">
316.                 <label>Tanggal          Kembali          <span
class="req">*</span></label>
317.                 <input type="date"
318.                     name="tgl_kembali"
319.                     id="tgl_kembali"
320.                     value="<?=
htmlspecialchars($data['tgl_kembali'] ?? '') ?>"
321.                     required>
322.                 <p class="hint" id="hint_kembali">Harus
≥ tanggal pinjam</p>
323.             </div>
324.         </div>
325.
326.         <button    type="submit"    class="btn-submit"><?=
$namaBtn ?></button>
327.     </form>
328. </div>
329. </div>
330.
331. <script>
332.     const          tglPinjamInput          =
document.getElementById('tgl_pinjam');

```

```

333.         const          tglKembaliInput          =
document.getElementById('tgl_kembali');
334.         const          hintKembali          =
document.getElementById('hint_kembali');
335.
336.         function updateMinKembali() {
337.             const tglPinjam = tglPinjamInput.value;
338.
339.             if (tglPinjam) {
340.                 tglKembaliInput.min = tglPinjam;
341.
342.                 if          (tglKembaliInput.value          &&
tglKembaliInput.value < tglPinjam) {
343.                     tglKembaliInput.value = tglPinjam;
344.                 }
345.
346.                 hintKembali.textContent = 'Minimal: ' +
formatTanggal(tglPinjam);
347.                 hintKembali.style.color = '#1e4d8c';
348.             }
349.         }
350.
351.         function formatTanggal(dateStr) {
352.             if (!dateStr) return '';
353.             const [y, m, d] = dateStr.split('-');
354.             const          bulan          =
['Jan', 'Feb', 'Mar', 'Apr', 'Mei', 'Jun', 'Jul', 'Agu', 'Sep', 'Okt',
'Nov', 'Des'];
355.             return    `${parseInt(d)}    ${bulan[parseInt(m)-1]}
${y}`;
356.         }
357.
358.         document.getElementById('formPeminjaman').addEventListener('
submit', function(e) {
359.             const tglPinjam = tglPinjamInput.value;
360.             const tglKembali = tglKembaliInput.value;
361.
362.             if (tglPinjam && tglKembali && tglKembali <
tglPinjam) {
363.                 e.preventDefault();
364.                 tglKembaliInput.classList.add('error');
365.                 hintKembali.textContent = '⚠ Tanggal kembali
tidak boleh sebelum tanggal pinjam!';
366.                 hintKembali.style.color = '#ef4444';
367.                 tglKembaliInput.focus();
368.             }
369.         });
370.
371.         tglKembaliInput.addEventListener('change', function() {
372.             const tglPinjam = tglPinjamInput.value;

```

```

373.         const tglKembali = this.value;
374.
375.         if (tglPinjam && tglKembali && tglKembali <
tglPinjam) {
376.             this.classList.add('error');
377.             hintKembali.textContent = '⚠ Tanggal kembali
tidak boleh sebelum tanggal pinjam!';
378.             hintKembali.style.color = '#ef4444';
379.         } else {
380.             this.classList.remove('error');
381.             hintKembali.textContent = tglPinjam ? 'Minimal:
' + formatTanggal(tglPinjam) : 'Harus ≥ tanggal pinjam';
382.             hintKembali.style.color = '#999';
383.         }
384.     });
385.
386.     tglPinjamInput.addEventListener('change',
updateMinKembali);
387.
388.     updateMinKembali();
389. </script>
390.
391. </body>
392. </html>

```

Tabel 11. Source Code Database.sql (perpustakaan)

```

1. CREATE DATABASE IF NOT EXISTS perpustakaan
2.     CHARACTER SET utf8
3.     COLLATE utf8_general_ci;
4.
5. USE perpustakaan;
6.
7. CREATE TABLE IF NOT EXISTS member (
8.     id_member          INT AUTO_INCREMENT PRIMARY KEY,
9.     nama_member        VARCHAR(250) NOT NULL,
10.    nomor_member       VARCHAR(15)  NOT NULL UNIQUE,
11.    alamat              TEXT,
12.    tgl_mendaftar      DATETIME      DEFAULT CURRENT_TIMESTAMP,
13.    tgl_terakhir_bayar DATE
14. );
15.
16. CREATE TABLE IF NOT EXISTS buku (
17.     id_buku            INT AUTO_INCREMENT PRIMARY KEY,
18.     judul_buku         VARCHAR(500) NOT NULL,
19.     penulis            VARCHAR(500) NOT NULL,
20.     penerbit           VARCHAR(250),
21.     tahun_terbit       INT
22. );
23.
24. CREATE TABLE IF NOT EXISTS peminjaman (
25.     id_peminjaman     INT AUTO_INCREMENT PRIMARY KEY,
26.     id_member          INT NOT NULL,

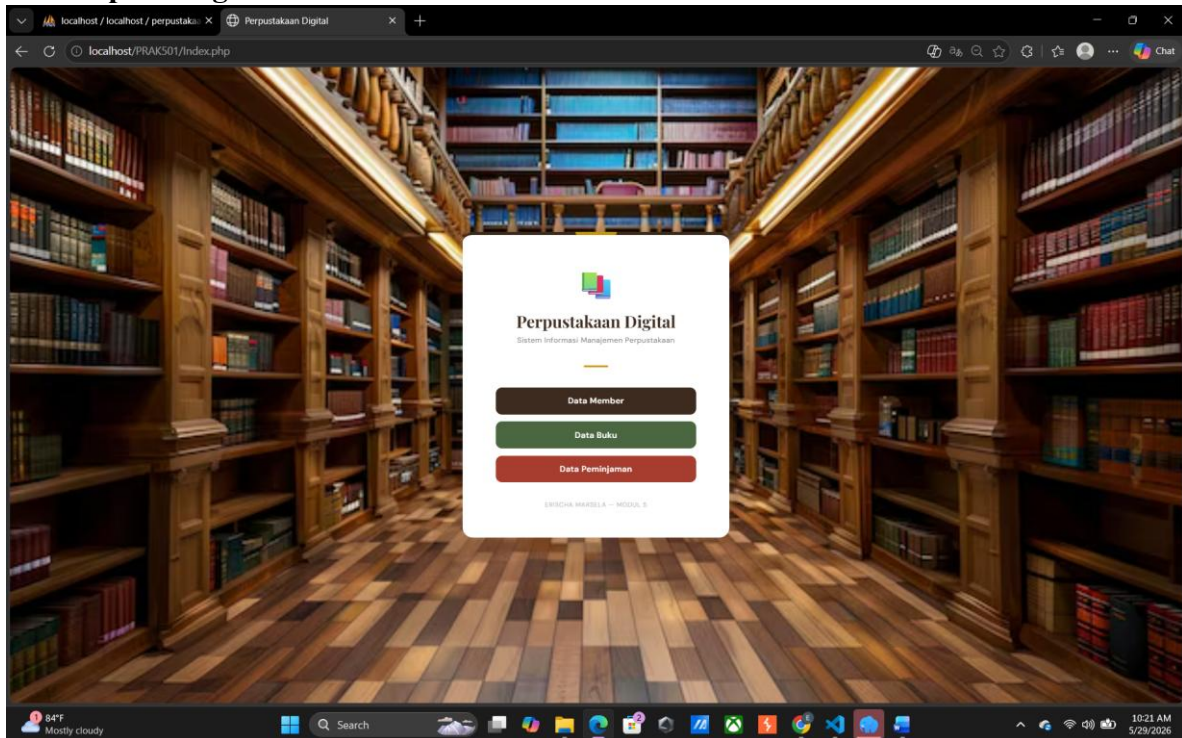
```

```

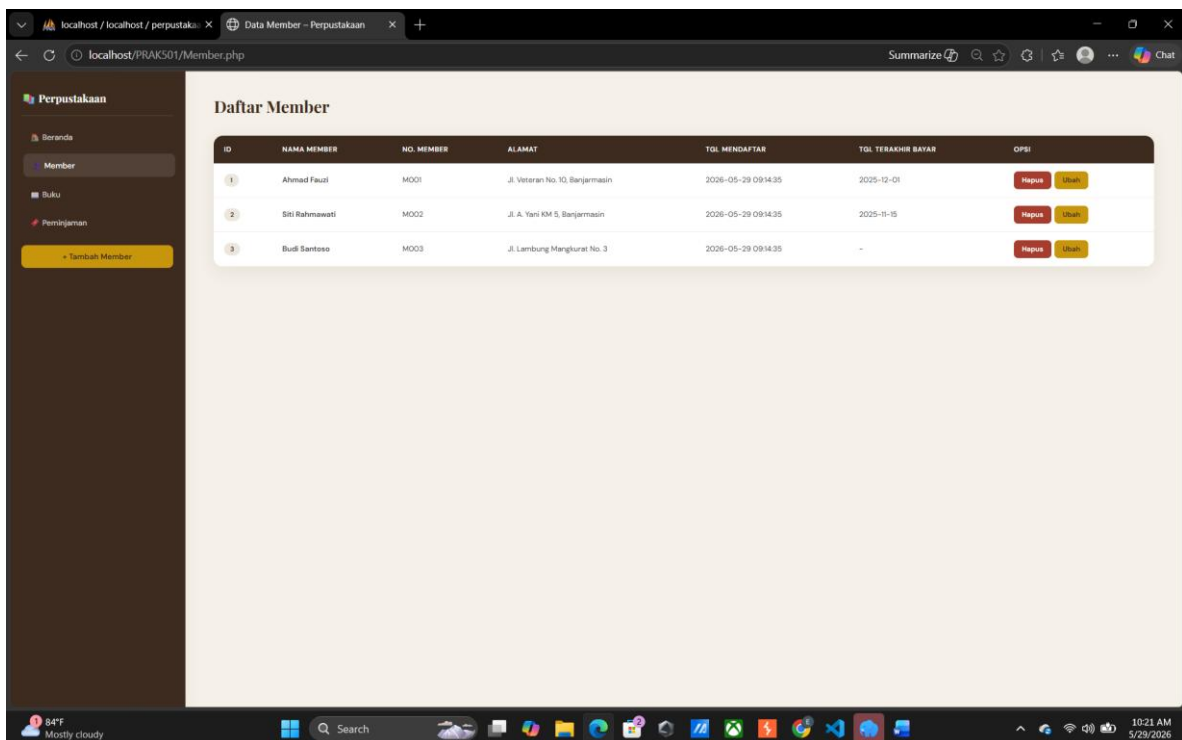
27.      id_buku          INT NOT NULL,
28.      tgl_pinjam       DATE NOT NULL,
29.      tgl_kembali      DATE NOT NULL,
30.      status           ENUM('aktif','selesai') NOT NULL DEFAULT
      'aktif',
31.      CONSTRAINT fk_peminjaman_member
32.          FOREIGN KEY (id_member) REFERENCES member(id_member)
33.          ON DELETE CASCADE ON UPDATE CASCADE,
34.      CONSTRAINT fk_peminjaman_buku
35.          FOREIGN KEY (id_buku) REFERENCES buku(id_buku)
36.          ON DELETE CASCADE ON UPDATE CASCADE
37. );
38.
39. INSERT INTO member (nama_member, nomor_member, alamat,
      tgl_mendaftar, tgl_terakhir_bayar) VALUES
40. ('Ahmad Fauzi', 'M001', 'Jl. Veteran No. 10,
      Banjarmasin', NOW(), '2025-12-01'),
41. ('Siti Rahmawati', 'M002', 'Jl. A. Yani KM 5, Banjarmasin',
      NOW(), '2025-11-15'),
42. ('Budi Santoso', 'M003', 'Jl. Lambung Mangkurat No. 3',
      NOW(), NULL);
43.
44. INSERT INTO buku (judul_buku, penulis, penerbit,
      tahun_terbit) VALUES
45. ('Negeri Para Bedebah', 'Tere Liye', 'Gramedia', 2012),
46. ('Negeri Di Ujung Tanduk', 'Tere Liye', 'Gramedia', 2013),
47. ('Bumi', 'Tere Liye', 'Gramedia', 2014),
48. ('Laskar Pelangi', 'Andrea Hirata', 'Bentang
      Pustaka', 2005),
49. ('Sang Pemimpi', 'Andrea Hirata', 'Bentang
      Pustaka', 2006);
50.
51. INSERT INTO peminjaman (id_member, id_buku, tgl_pinjam,
      tgl_kembali) VALUES
52. (1, 1, '2025-01-10', '2025-01-24'),
53. (2, 3, '2025-01-15', '2025-01-29'),
54. (3, 5, '2025-02-01', '2025-02-15');

```


B. Output Program



Gambar 2. Dashboard Page



Gambar 3. Daftar Member Page

Tambah Data Member
Lengkapi semua field yang wajib diisi (*)

Nama Member *
Masukkan nama lengkap

Nomor Member *
Contoh: MO01

Alamat
Masukkan alamat lengkap

Tanggal Mendaftar *
05/29/2026 03:22 AM

Tanggal Terakhir Bayar
mm/dd/yyyy

Daftarkan Member

Gambar 4. Form Member Page

Perpustakaan

- Beranda
- Member
- Buku**
- Peminjaman
- Tambah Buku

Daftar Buku

ID	JUDUL BUKU	PENULIS	PENERBIT	TAHUN TERBIT	OPSI
1	Negeri Para Bidadari	Tere Liye	Gramedia	2012	Hapus Ubah
2	Negeri Di Ujung Tanduk	Tere Liye	Gramedia	2013	Hapus Ubah
3	Bumi	Tere Liye	Gramedia	2014	Hapus Ubah
4	Laskar Pelangi	Andrea Hirata	Bersang Pustaka	2005	Hapus Ubah
5	Sang Pemimpi	Andrea Hirata	Bersang Pustaka	2006	Hapus Ubah

Gambar 5. Daftar Buku Page

Tambah Data Buku
Isi informasi buku dengan lengkap dan benar.

Judul Buku *
Masukkan judul buku

Penulis *
Nama penulis

Penerbit
Nama penerbit

Tahun Terbit
Contoh: 2023

Tambahkan Buku

Gambar 6. Form Buku Page

Data Peminjaman

ID	NAMA MEMBER	NO. MEMBER	JUDUL BUKU	TGL PINJAM	TGL KEMBALI	STATUS	OPSI
1	Ahmad Fauzi	MO01	Negeri Para Bidebah	2025-01-10	2026-05-29	Selesai	Selesai Hapus
2	Siti Rahmawati	MO02	Bumi	2025-01-15	2025-01-29	Terlambat	Selesai Ubah Hapus
3	Budi Santoso	MO03	Sang Pemimpi	2025-02-01	2025-02-15	Terlambat	Selesai Ubah Hapus

Gambar 7. Data Peminjaman Page

The image shows a web browser window with the title 'Perpustakaan Digital'. The address bar displays 'localhost/PRAKS01/FormPeminjaman.php'. The main content area features a form titled 'Tambah Data Peminjaman' with the subtitle 'Pilih member dan buku, kemudian tentukan tanggal peminjaman.' The form contains four input fields: 'Nama Member' with a dropdown menu showing 'Pilih Member', 'Judul Buku' with a dropdown menu showing 'Pilih Buku', 'Tanggal Pinjam' with a date picker showing '05/29/2026', and 'Tanggal Kembali' with a date picker showing 'mm/dd/yyyy' and a note 'Minimal 20 Mei 2026'. A green button labeled 'Catat Peminjaman' is positioned below the date fields. The browser's taskbar at the bottom shows the system clock as 10:22 AM on 5/29/2026.

Gambar 8. Form Peminjaman Page

C. Pembahasan

Koneksi.php mendefinisikan konstanta konfigurasi penting seperti nama host, username, kata sandi, dan nama basis data perpustakaan, sekaligus menyediakan fungsi khusus bernama koneksiDatabase() untuk menghubungkan aplikasi ke MySQL. Koneksi ini kemudian dimanfaatkan oleh berkas Model.php yang bertindak sebagai pusat kendali data, di mana di dalamnya terdapat berbagai fungsi operasional seperti getAllMember(), insertBuku(), hingga deletePeminjaman(). Secara mendetail namun ringkas, setiap fungsi di dalam berkas ini akan menjalankan perintah SQL standar dan secara aman menerapkan fungsi mysqli_real_escape_string untuk mencegah gangguan error atau serangan injeksi kode saat data dimasukkan ke dalam basis data.

Sementara itu, bagian tampilan atau antarmuka pengguna dikendalikan oleh berkas seperti Index.php, Member.php, Buku.php, Peminjaman.php, serta berkas formulir (Form) terkait. Berkas Index.php berfungsi sebagai halaman utama atau dasbor yang menyajikan menu navigasi untuk mengarahkan pengguna ke pengelolaan data member, buku, atau transaksi peminjaman. Ketika pengguna melakukan sebuah aksi di halaman antarmuka misalnya menekan tombol hapus atau mengisi formulir tambah data halaman tersebut akan menangkap input pengguna, meneruskannya ke fungsi yang tepat di dalam Model.php untuk diproses ke database, dan akhirnya memuat ulang halaman dengan menampilkan pesan notifikasi bahwa operasi telah berhasil dilakukan.

Tautan Git

<https://github.com/ERISCHA24/PRAKTIKUMWEB2>